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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	
Student Name	Alluri Eswari Kali Priya
Roll Number	2520030492
Branch	CSE
Course Outcome	CO2

Case Study Topic: Foxconn Factory Floor – Segment Tree for Sensor Range-Max Alerts

Work Done Summary:
In this case study, a Segment Tree was implemented to process temperature sensor readings efficiently. The system supports range maximum queries and point updates for real-time monitoring applications.

Details:

- • Created an array to store sensor temperature readings.
- • Constructed a Segment Tree using recursive tree building.
- • Stored maximum values in internal tree nodes.
- • Implemented range maximum query functionality.
- • Handled complete, partial, and no-overlap cases.
- • Performed point update operations on sensor readings.
- • Updated affected tree nodes after modification.
- • Displayed query results before and after updates.

Result: Screenshots/Output

Output

```
FACTORY SENSOR ALERT SYSTEM
---------------------------
Maximum Temperature [3..7] = 82
After Update (75 -> 88)
Maximum Temperature [2..6] = 88

=== Code Execution Successful ===
```

Observations:

Segment Tree was successfully constructed from sensor data.

- Range maximum queries returned correct results.
- Point updates propagated correctly to parent nodes.
- Query operations required fewer comparisons than linear search.
- Update operations were completed efficiently.
- The system supported real-time sensor monitoring.

GitHub Link: https://github.com/eswarikalipriya-boop/DSA.git

Student Signature	Faculty Signature